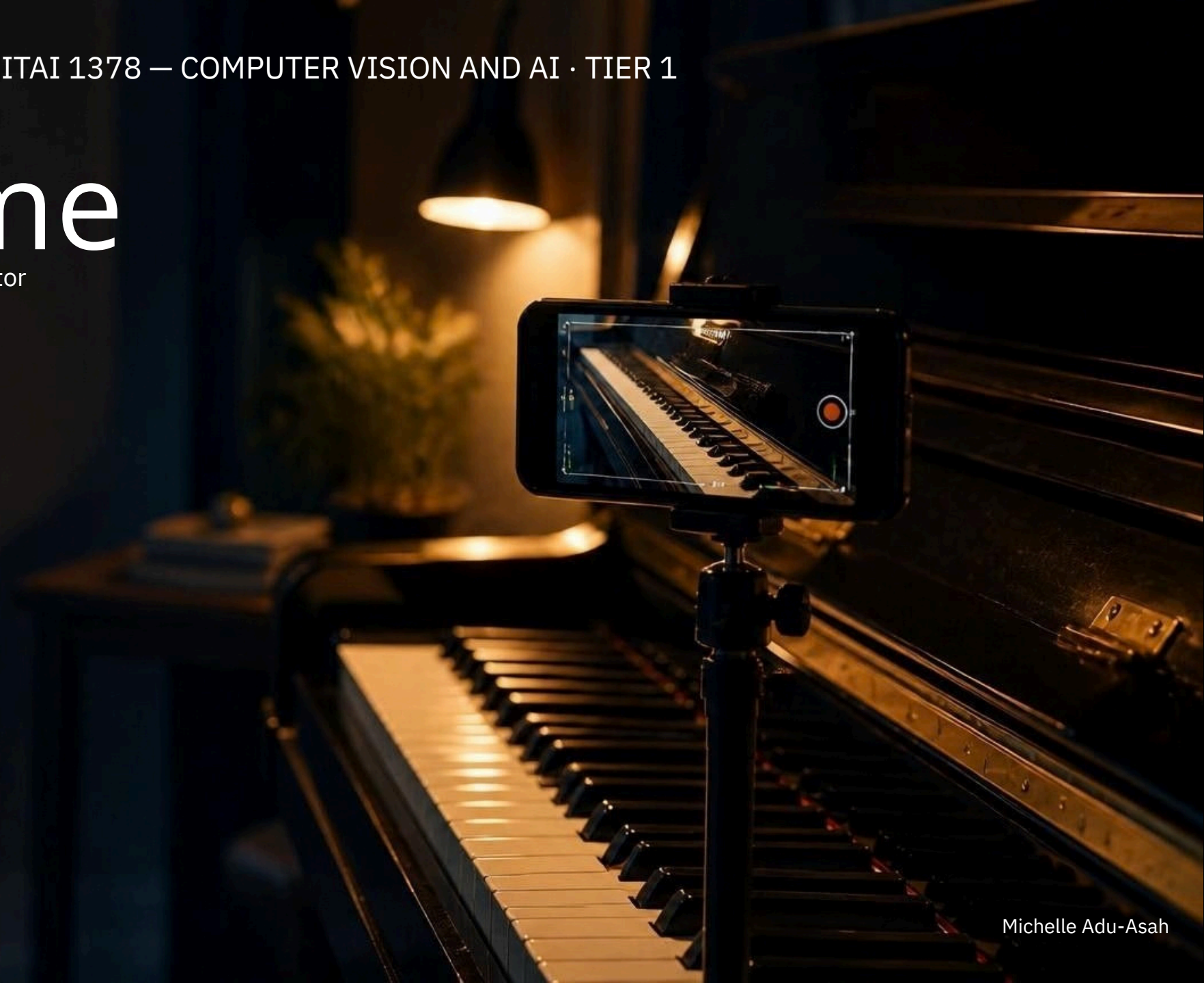


KeyFrame

Piano Key Press Detector

Using computer vision to detect
piano key presses from video.





The Problem

- I teach piano to kids during the summer at my church, and beginners can struggle to know whether they are pressing the correct keys.
- Teachers cannot watch every student practice all the time.
- A simple visual check could help students practice with more confidence, without replacing a teacher.
- Who it is for: beginner piano students, piano teachers, and community music programs, and students practicing at home by themselves.

THE IDEA

A phone camera can turn a key press into a note

My goal: watch a piano keyboard, detect which key is pressed, and show its note.

01

Phone camera

02

Keyboard video

03

Computer vision

04

Pressed key
detected

05

Note
shown

Example

Student presses C → system detects
C → "C4"

The first prototype will focus on about 5–7 keys to keep the project manageable.

I will start with video changes, not a trained AI model

01

Record

Keyboard video on my smartphone.

02

Read

Process the video frame by frame.

03

Mark

Identify regions for selected keys.

04

Compare

Look for visual changes that indicate a press.

05

Output

Identify the pressed key and show its note.

Tools Python · OpenCV · Google Colab · smartphone camera

OpenCV gives me practical image and video tools. A small key range keeps the first version realistic for my experience and timeline.

DATA PLAN

My data will come from short controlled recordings

Source: self-collected smartphone video.

Estimated set: about 5–10 short recordings, with multiple examples of each selected key.

Initial labels: C, D, E, F, and G; more keys only if the first version works well.

Verification: Cubase MIDI recordings can provide a reference for the exact notes played.



HOW I'LL MEASURE SUCCESS

Success means getting the note right, quickly

PRIMARY GOAL · KEY-DETECTION ACCURACY

90%

If I test 100 key presses, the system should correctly identify at least 90.

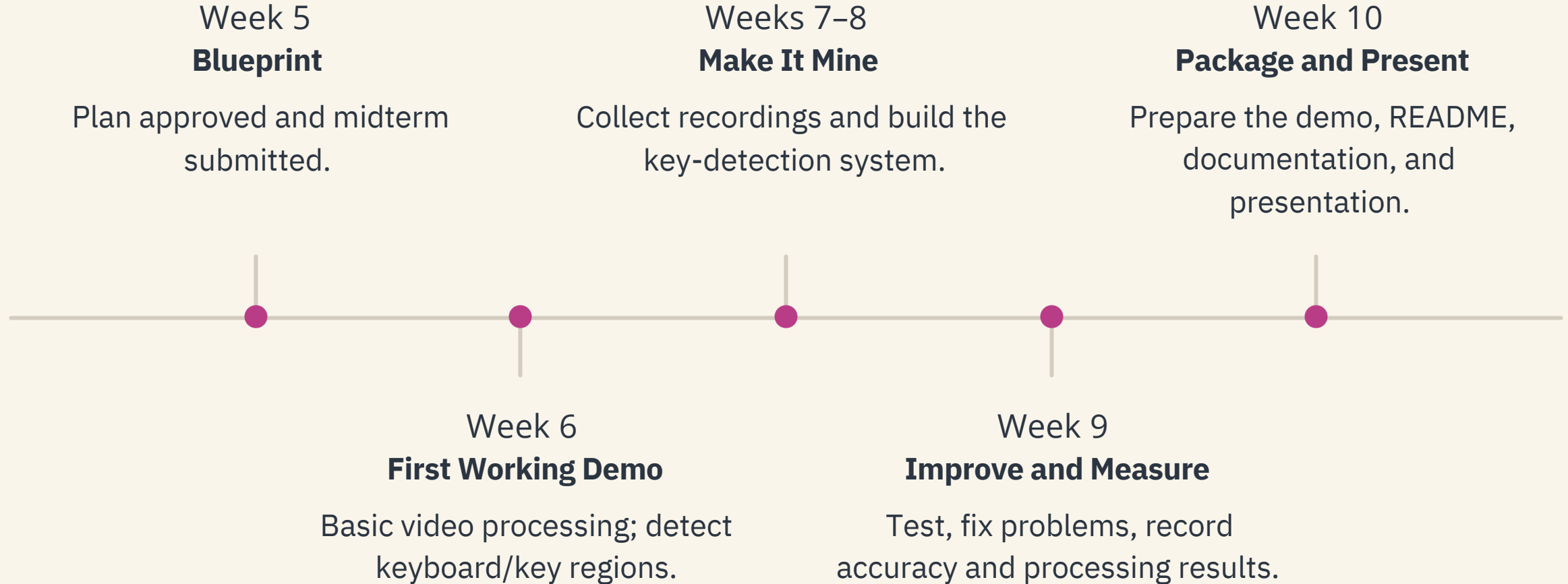
SECONDARY GOAL

<1 sec

Processing time per detected key

Testing: use separate recordings that were not used while developing the system.

From idea to prototype



Risks and resources

Keys look too similar

Risk · Pressed and unpressed keys may look too similar in some frames.

Plan B · Fixed camera position, better lighting, and fewer keys.

Real-time detection is hard

Risk · Real-time detection may be harder than expected.

Plan B · Process a prerecorded smartphone video instead of requiring live detection.

Resources

Smartphone · my midi keyboard · Python · OpenCV · Google Colab · GitHub · Cubase

COST

\$0

I already have the needed equipment, subscriptions and can use free software.